

PLANETARY GEOPHYSICS

Isotopic evidence of long-lived volcanism on Io

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Jupiter's moon Io hosts extensive volcanism, driven by tidal heating. The isotopic composition of Io's inventory of volatile chemical elements, including sulfur and chlorine, reflects its outgassing and mass-loss history and thus records information about its evolution. We used submillimeter observations of Io's atmosphere to measure sulfur isotopes in gaseous sulfur dioxide and sulfur monoxide, and chlorine isotopes in gaseous sodium chloride and potassium chloride. We find $^{34}\text{S}/^{32}\text{S} = 0.0595 \pm 0.0038$ (equivalent to $\delta^{34}\text{S} = +347 \pm 86\text{‰}$), which is highly enriched compared to average Solar System values and indicates that Io has lost 94 to 99% of its available sulfur. Our measurement of $^{37}\text{Cl}/^{35}\text{Cl} = 0.403 \pm 0.028$ ($\delta^{37}\text{Cl} = +263 \pm 88\text{‰}$) shows that chlorine is similarly enriched. These results indicate that Io has been volcanically active for most (or all) of its history, with potentially higher outgassing and mass-loss rates at earlier times.

Widespread volcanic activity on Jupiter's moon Io is powered by tidal heating of its interior, which arises from Io's orbital resonance with the neighboring moons Europa and Ganymede. Models of the formation of Jupiter's large moons show that Io, Europa, and Ganymede were probably captured into the resonance during their formation process (1, 2). If so, Io and Europa have experienced strong tidal heating for the entire 4.57-billion year (Gyr) history of the Solar System, implying that Io has been volcanically active (either continuously or cyclically) over the same period (3).

Io's current volcanic activity resurfaces the moon at a rate of 0.1 to 1.0 cm year⁻¹ (4). This has erased all impact craters from its surface (5), leaving a geological record of only the most recent million years of its history. However, isotopic abundances could record the history of volcanism on Io: If Io's current rates of mass loss [1000 to 3000 kg s⁻¹ (6)] and outgassing from its interior to its atmosphere have been sustained for billions of years, its reservoirs of volatile elements should be highly enriched in heavy stable isotopes, because atmospheric escape processes generally favor the loss of lighter isotopes. Stable isotope measurements of volatile elements, such as sulfur and chlorine, could provide information on the history of volcanism on Io.

Submillimeter observations of Io

We used the Atacama Large Millimeter/submillimeter Array (ALMA) to observe gases in Io's atmosphere (7). Io is tidally locked to Jupiter; the hemispheres facing into and away from its direction of motion (its leading and trailing hemispheres) were observed on 2022 May 24 and 2022 May 18, respectively (Universal Time, UT). We used ALMA's Band 8 receivers to cover the frequency range of 416 to 432 GHz (~0.7 mm wavelength) in 13 spectral windows. This frequency range was chosen to cover multiple rotational transitions of SO₂, SO, NaCl, KCl, and their isotopologues, with the goal of determining the $^{34}\text{S}/^{32}\text{S}$ and $^{37}\text{Cl}/^{35}\text{Cl}$ ratios. The data were processed (7) to produce a calibrated spectral data cube, with dimensions of right ascension (RA), declination (Dec), and frequency, with Io's thermal emission continuum subtracted. The data have a spectral resolution of 244 kHz (170 m s⁻¹) and an angular resolution of ~0.28 arc sec (") (equivalent to a spatial resolution of ~1000 km at the distance of Io at the time of our observations). The data cubes and extracted images for each species have a pixel scale of 0".03, such that the spatial resolution is sampled with ~10 pixels per resolution element (7).

Figure 1 shows the spatial distribution of each of the four gas species that we targeted. SO₂ and SO are concentrated in the low-to-mid latitudes, with the strongest emission close to the limb, where the atmospheric path length is longest. By contrast, NaCl and KCl are confined to a few localized points, which we interpret as volcanic plumes (locations listed in table S4).

From the data cubes, we extracted disk-integrated spectra around the targeted emission lines (Figs. 2 and 3), using an aperture defined by all pixels in which the continuum emission is at least 5% of the peak continuum emission; this produces an aperture radius that is roughly 1.3× Io's radius. We quantify the noise in the spectrum for each of the 13

spectral windows independently, calculating it as the standard deviation of the disk-integrated spectrum in line-free regions.

Atmospheric modeling

We used a radiative transfer model of Io's tenuous atmosphere (7–9) to determine the $^{34}\text{S}/^{32}\text{S}$ ratio, by fitting the observed emission lines of SO₂. SO₂ makes up ~90 to 97% of Io's atmosphere (10), so the abundances of all other species in the model are calculated as mixing ratios relative to SO₂. To determine the $^{34}\text{S}/^{32}\text{S}$ ratio, we jointly fitted two lines of $^{32}\text{SO}_2$ and four lines of $^{34}\text{SO}_2$. The SO lines were not used in the model fitting, because of the low signal-to-noise ratio (SNR) of the ^{34}SO lines. The free parameters in the model are the SO₂ column density, the $^{34}\text{S}/^{32}\text{S}$ ratio, and the gas temperature. The gas temperature is constrained because the dataset includes both high- and low-excitation lines. The selection of which $^{34}\text{SO}_2$ and $^{32}\text{SO}_2$ lines to observe was designed to ensure that they were sensitive to the same altitudes (7); therefore, the derived isotopic ratio is not strongly sensitive to the assumed atmospheric temperature profile.

The $^{37}\text{Cl}/^{35}\text{Cl}$ ratio was determined with the same procedure, by jointly fitting the observed NaCl and KCl lines for both chlorine isotopes (five lines total). The free parameters are the NaCl and KCl column densities (relative to SO₂), the gas temperature, the gas fractional surface coverage (the ratio of the emitting area of the gas to the projected surface area of Io), the $^{37}\text{Cl}/^{35}\text{Cl}$ ratio, and the line-of-sight gas velocity (relative to Io's velocity and rotation). Gas velocity is a free parameter in the chlorine model because the NaCl and KCl lines are all frequency-shifted from Io's rest frame, with each species shifted in the same way within each observation, indicating bulk motion of the chlorine-bearing gas relative to Io's rotation. This parameter was not necessary for fitting the SO₂ lines.

The observed lines are compared to the best-fitting models in Figs. 2 and 3; the model parameters are listed in Table 1. The SO data are also shown in Fig. 2, which compares them to the model derived from SO₂; we find that the SO data are consistent with the same isotope ratio as derived from SO₂. On Io, SO is produced by photodissociation of SO₂ (11), so we expect its $^{34}\text{S}/^{32}\text{S}$ ratio to only differ from that of SO₂ by a factor of 1.009 (12), which is smaller than the uncertainties. The best-fitting SO/SO₂ ratios are 3 and 5% for the leading and trailing hemispheres, respectively, which are within the range of 3 to 10% found by previous studies (10, 13).

The best-fitting gas temperatures for NaCl and KCl (assumed to be identical) are 774 ± 66 and 682 ± 56 K for the leading and trailing hemispheres, respectively (all uncertainties are

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